District-Level Social Deprivation in India

A Multidimensional Analysis using NFHS-5 (2019-21)

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1 Executive Summary

This report constructs a **Composite Deprivation Index** for 707 districts across India using data from the National Family Health Survey (NFHS-5, 2019-21). The index captures deprivation across three dimensions — Health, Education, and Living Standards — using 13 indicators.

Key findings:

- **Health is the most deprived dimension nationally**, with an average score of 41.5 out of 100.
- **Bihar and Tripura** are the most deprived states; **Kerala and Mizoram** are the least deprived.
- **Education deprivation is the strongest predictor** of overall deprivation, suggesting it should be a primary policy lever.
- **Gujarat** ranks among the most deprived states despite being one of India's wealthier states by GDP — highlighting the limits of economic growth as a measure of human development.
- Significant **within-state inequality** exists: in Madhya Pradesh, the gap between the most and least deprived district spans 26 index points.

2 Data and Methodology

2.1 Data Source

This analysis uses the publicly available **NFHS-5 district-level factsheet data** released by the Ministry of Health and Family Welfare, Government of India. The dataset covers 707 districts across all states and union territories.

2.2 Indicator Selection

Thirteen indicators were selected across three dimensions:

Indicators used in the Deprivation Index

Dimension	Indicator	Direction
Living Standards	Access to electricity	Higher = better
Living Standards	Improved drinking water source	Higher = better
Living Standards	Improved sanitation facility	Higher = better
Living Standards	Clean cooking fuel	Higher = better
Living Standards	Health insurance coverage	Higher = better
Education	Women's literacy rate	Higher = better
Education	Women with 10+ years schooling	Higher = better
Education	Child marriage rate	Lower = better
Health	Institutional births	Higher = better
Health	Full vaccination coverage	Higher = better
Health	Child stunting rate	Lower = better
Health	Child underweight rate	Lower = better
Health	Women anaemia rate	Lower = better

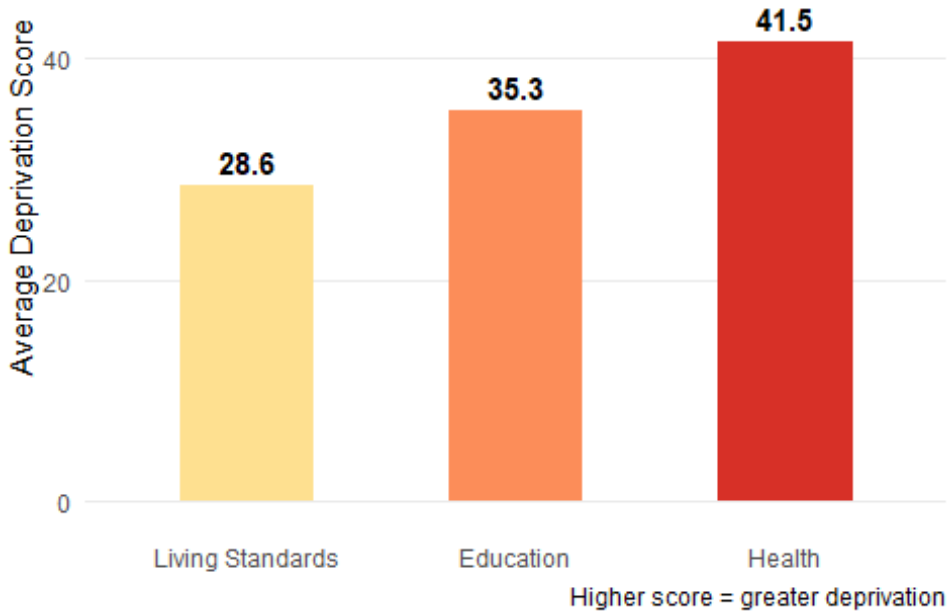
2.3 Index Construction

Each indicator was converted to a deprivation score (0-100) where higher values always indicate greater deprivation. For "higher = better" indicators, the score was computed as `100 - original value`. Dimension scores were computed as simple averages of their constituent indicators. The final Composite Deprivation Index is an equal-weighted average of the three dimension scores.

This approach follows the **Alkire-Foster methodology** used by NITI Aayog and the Oxford Poverty and Human Development Initiative (OPHI).

3 National Findings

3.1 Deprivation by Dimension



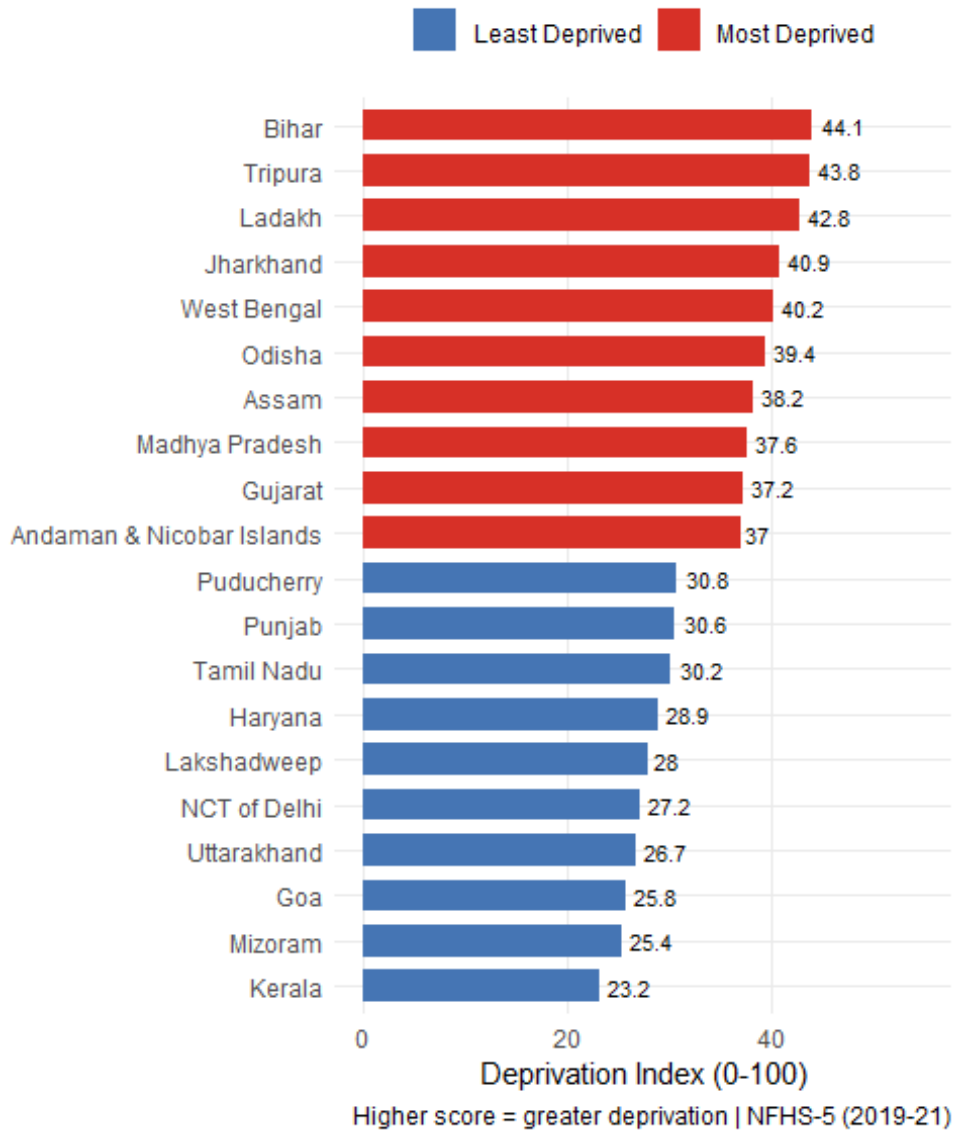
Average deprivation score by dimension across 707 districts

Health emerges as the most deprived dimension nationally (41.5), followed by Living Standards (28.6) and Education (35.3). However, correlation analysis reveals that **Education is the strongest predictor of overall deprivation** (r = 0.68), suggesting that districts with poor educational outcomes tend to struggle across all dimensions simultaneously.

Notably, Health deprivation shows a weak negative correlation with both Education (-0.20) and Living Standards (-0.19), indicating that **health deprivation operates as a structurally independent problem** — districts can have relatively good infrastructure and literacy yet still suffer from high stunting and anaemia rates. This points to failures in nutrition delivery and primary healthcare that transcend income levels.

4 State-Level Analysis

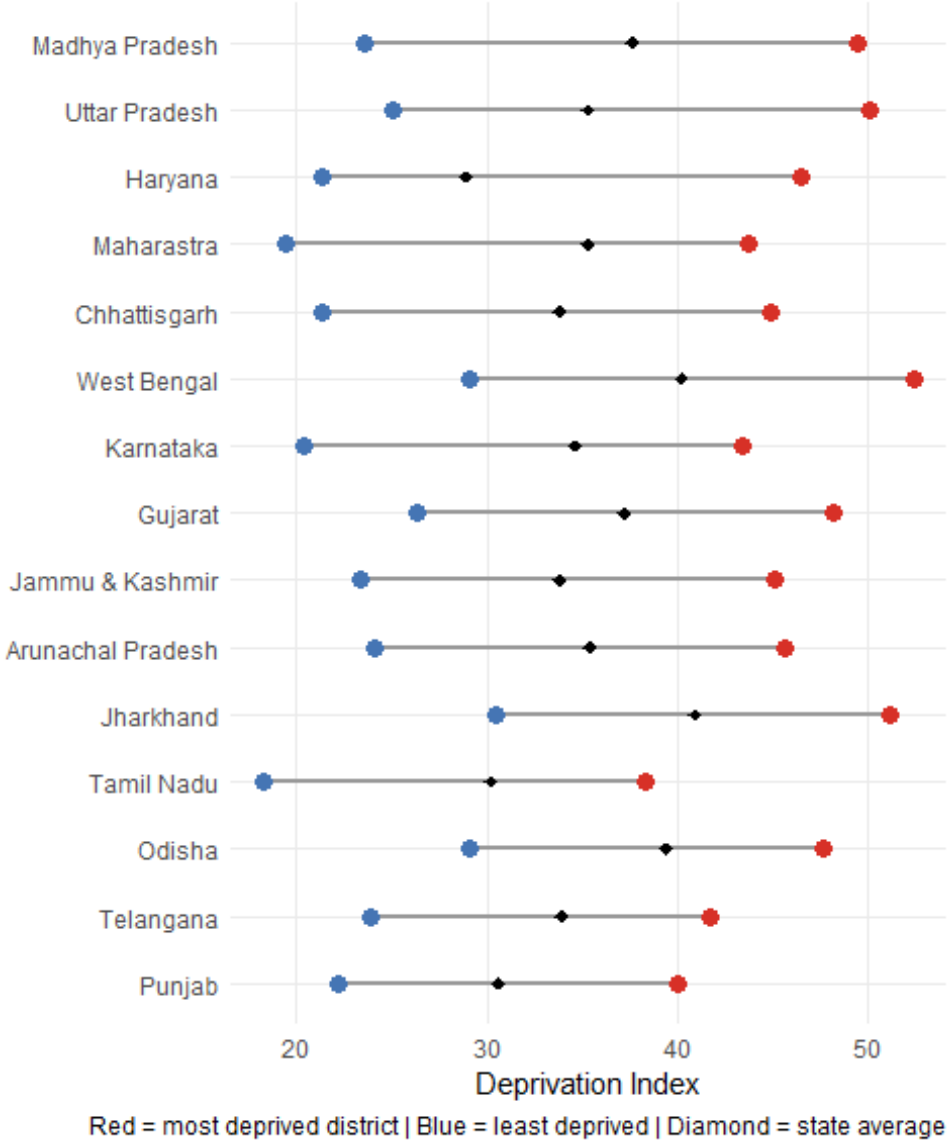
4.1 Most and Least Deprived States



Top and bottom 10 states by composite deprivation index

Bihar (44.1) and Tripura (43.8) are the most deprived states. Kerala (23.2) is the least deprived, with an education deprivation score of just 10.2 — the lowest of any state in India. **Ladakh (42.8) is a notable outlier**: its high overall deprivation is driven almost entirely by health (67.5), reflecting extreme geographic remoteness rather than conventional poverty. **Gujarat (37.2) ranks among the most deprived despite being one of India's top states by GDP per capita** — a finding that underscores the disconnect between economic growth and human development outcomes.

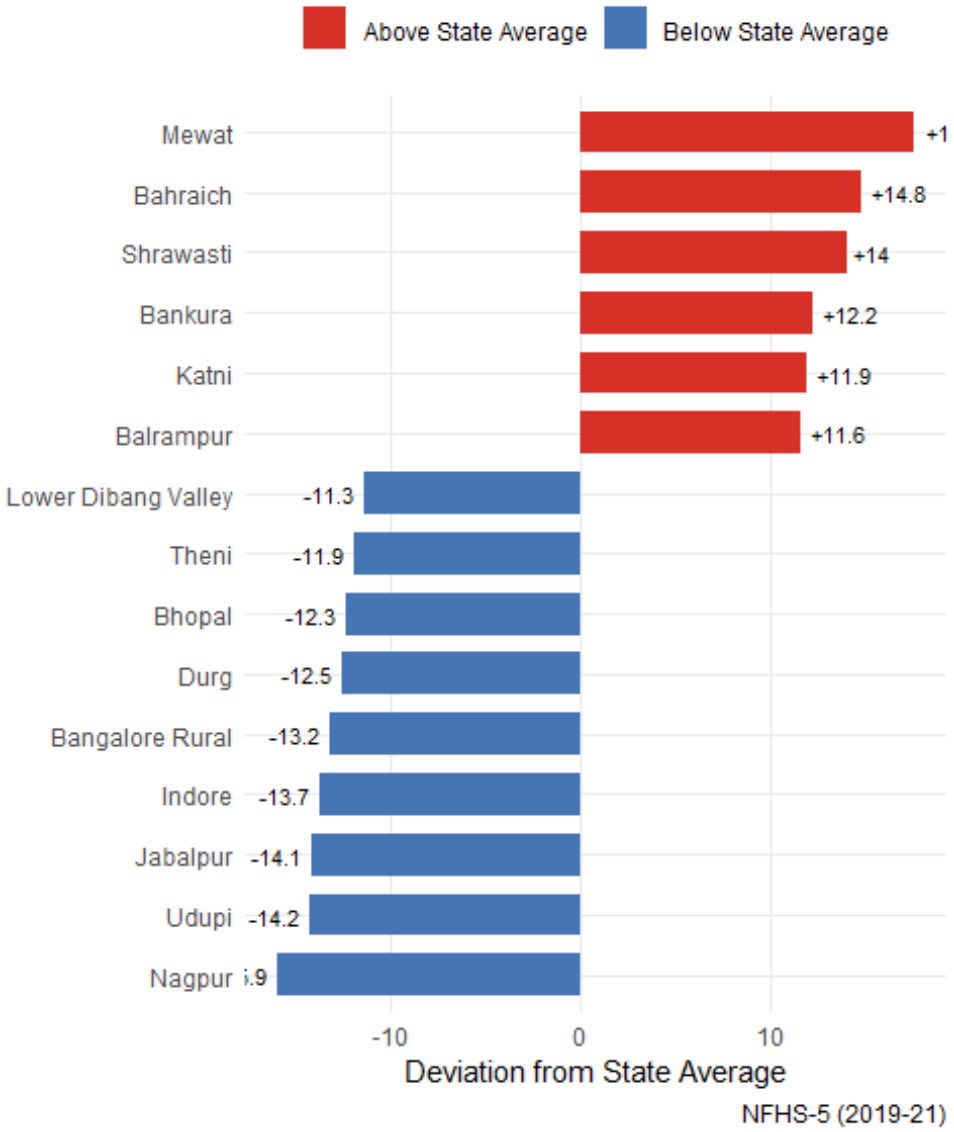
4.2 Within-State Inequality



Range of deprivation between best and worst district within each state

State averages mask significant internal disparities. Madhya Pradesh has the widest district-level range (26 points), followed by Haryana (25.1) and Maharashtra (24.3). These states require **geographically targeted interventions** rather than uniform state-level policies.

5 District Spotlight — Outliers



Districts with largest deviation from their state average deprivation score

Three districts warrant particular attention:

- **Mewat, Haryana (+17.6)**: The most extreme outlier in India — deprived across all three dimensions in an otherwise moderately developed state. Persistent underdevelopment driven by low female literacy and poor maternal health outcomes.
- **Bankura, West Bengal (+12.2)**: High health deprivation (67.6) despite moderate living standards — consistent with the finding that health deprivation can be structurally independent of infrastructure access.

- **Nagpur, Maharashtra (-15.9)**: The strongest positive outlier — a major urban centre dramatically outperforming its state average, illustrating the urban concentration of development gains.

6 Conclusion

This analysis of 707 Indian districts reveals that deprivation in India is **multidimensional, geographically uneven, and structurally complex**. Three policy implications emerge:

1. Health requires a standalone policy focus. Its weak correlation with other dimensions means that improving infrastructure and education alone will not automatically reduce stunting, anaemia, or maternal health gaps. Dedicated nutrition and primary healthcare programmes are needed.

2. Education is the highest-leverage intervention. As the strongest predictor of overall deprivation, improvements in female literacy and schooling are likely to generate the broadest cross-dimensional benefits.

3. State-level policies are insufficient. The wide within-state variation in states like Madhya Pradesh and Haryana demands district-level targeting. Mewat's extreme deprivation within prosperous Haryana exemplifies how state averages can mask acute localised need.

Data Source: Ministry of Health and Family Welfare, Government of India. National Family Health Survey (NFHS-5), 2019-21. District Fact Sheets.

Index methodology follows the Alkire-Foster framework as applied by NITI Aayog's National Multidimensional Poverty Index (2021).